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Technical University
of Applied Sciences
Würzburg-Schweinfurt

Master's in Artificial Intelligence

Project Module

Winter Semester 2026/2027

Catalog of Project Topics

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Conversion of Air Traffic Control Audio into Dynamic Controls

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[15:58:13Z] Pilot: "Metro Approach, Skyline 482, descending through one-two thousand for eight thousand, information Bravo, requesting ILS runway 27."

[15:58:21Z] Air Traffic Control: "Skyline 482, Metro Approach, good afternoon. Expect ILS runway 27. Turn right heading two-four-zero, descend and maintain six thousand."

[15:58:28Z] Pilot: "Right heading two-four-zero, descend and maintain six thousand, Skyline 482."

[15:58:34Z] Air Traffic Control: "Skyline 482, reduce speed to two-one-zero knots."

[15:58:39Z] Pilot: "Speed two-one-zero, Skyline 482.

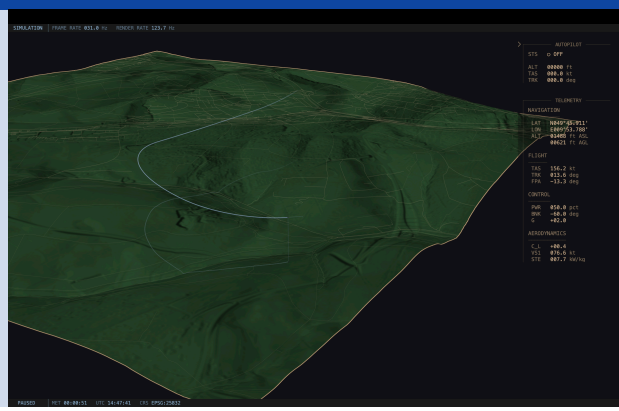


Figure 1: (**Left**) Communication between pilot and air traffic control. (**Right**) Instruction execution in simulation environment.

Context

Air traffic control (ATC) keeps our skies organized, efficient, and safe. Before an aircraft is allowed to enter the airspace surrounding an airport, the pilot must establish two-way radio communication with the airport tower and state their intentions, such as "inbound for landing." The controller in the tower then issues a series of instructions, including adjustments to speed, heading, and altitude, that the aircraft is required to follow; see Figure 1 for a typical exchange.

Motivation

Radio communication relies on analog signals and is therefore inherently noisy and non-parallel: only one speaker can transmit at a time. As a result, air traffic controllers must issue time-critical instructions to many aircraft in rapid succession, often over imperfect audio channels. Pilots receive these instructions during one of the most workload-intensive phases of flight, as they prepare the aircraft for landing. When pilots hear their aircraft identifier, they must interrupt their workflow, parse the instruction, and translate it into action, either manually or through the autopilot system. This process is error-prone: instructions may be misunderstood, misremembered, or incorrectly entered, especially under cognitive load. In several aviation accidents and incidents, such breakdowns in communication and workload management have contributed to fatal outcomes.

Project

The goal of this project is to shift the pilot from the active ATC communication loop into a supervisory role. To this end, a generative model, such as an LLM, will be trained to translate ATC audio instructions into aircraft control actions, such as adjustments to thrust, bank, and load factor, that satisfy the requested changes in speed, heading, or altitude. The model will read back the parsed instruction to ATC, after which the pilot will provide final confirmation to preserve safety and authority. The resulting system will be integrated into an existing simulation environment as an audio-controlled autopilot interface and evaluated using standardized industry metrics.

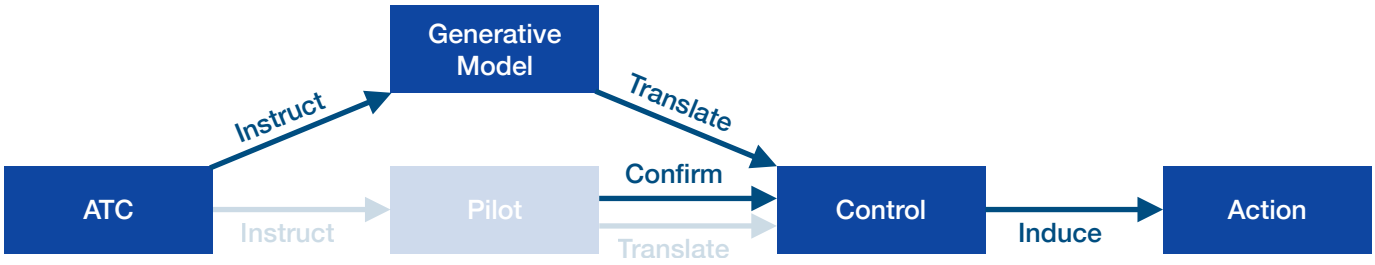
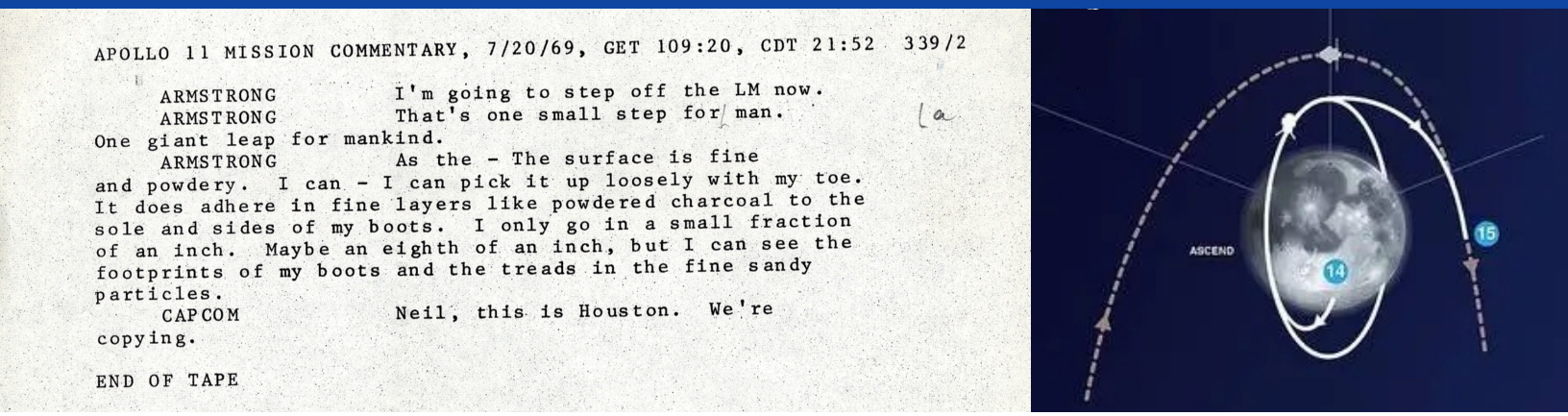


Figure 2: The role of the pilot in ATC communications shifts from active ( ) to passive ( ).

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It's Not Rocket Science: Generative Modeling of Moon Landing Dynamics

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APOLLO 11 MISSION COMMENTARY, 7/20/69, GET 109:20, CDT 21:52 339/2

ARMSTRONG I'm going to step off the LM now.
ARMSTRONG That's one small step for man.
One giant leap for mankind.
ARMSTRONG As the - The surface is fine and powdery. I can - I can pick it up loosely with my toe. It does adhere in fine layers like powdered charcoal to the sole and sides of my boots. I only go in a small fraction of an inch. Maybe an eighth of an inch, but I can see the footprints of my boots and the treads in the fine sandy particles.
CAPCOM Neil, this is Houston. We're copying.

END OF TAPE

Figure 1: Excerpt of official Apollo 11 transcript (**left**) and schematic illustration of lunar orbit rendezvous (**right**).

Context

Lunar orbit rendezvous was a central idea behind NASA's Apollo program and is receiving renewed interest in light of international plans for a crewed Moon landing in 2028. Rather than landing the entire spacecraft on the Moon, Apollo used a small Lunar Module (LM) that descended to the surface while the Command Module (CM) remained in lunar orbit. After completion of surface operations, the Lunar Module had to launch from the surface and perform a series of maneuvers to rendezvous with the Command Module in space.

Motivation

Classical astrodynamics provides equations and solvers for this problem, including linearized rendezvous models, numerical integration frameworks, and constrained trajectory optimization. However, many of these standard approaches are fully deterministic: given an objective function and an initial condition, they return a single optimal trajectory. Recovering alternative solutions often requires modifying the objective, rerunning the optimization, or hand-designing maneuvers and conditions. This limits their usefulness when the mission designer wants to understand the distribution of feasible strategies, such as faster high-fuel transfers, slower low-fuel arcs, safer approach geometries, or time-critical emergency returns. Standard solvers are also known to be computationally expensive, especially when embedded inside iterative planning pipelines or layered uncertainty analyses.

Project

This project explores the suitability of generative models as neural surrogates for lunar rendezvous planning. The goal is to develop efficient approximations that allow mission designers to sample diverse trajectories from a multimodal distribution of feasible strategies. To this end, you will develop a rendezvous simulator, generate training data from classical dynamics and traditional solvers, and use these data to optimize modern ML-architectures while preserving the underlying physics. The project consists of three phases:

- **Rendezvous from Existing Orbit.** The Lunar Module starts in a different nearby lunar orbit and must perform correction maneuvers to match the Command Module's relative position and velocity.
- **Ascent into Orbit.** The Lunar Module starts on the lunar surface and uses thrust control to reach a stable insertion orbit.
- **End-to-end lunar orbit rendezvous.** The Lunar Module starts on the lunar surface, reaches a suitable orbit, and then performs rendezvous maneuvers until it matches the Command Module's position and velocity.

The learned models will be rolled out in the simulator and compared to classical baselines based on evaluation criteria like terminal miss distance, residual relative velocity, constraint adherence, and trajectory diversity.

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Predicting Financial Markets through Competing Interactions

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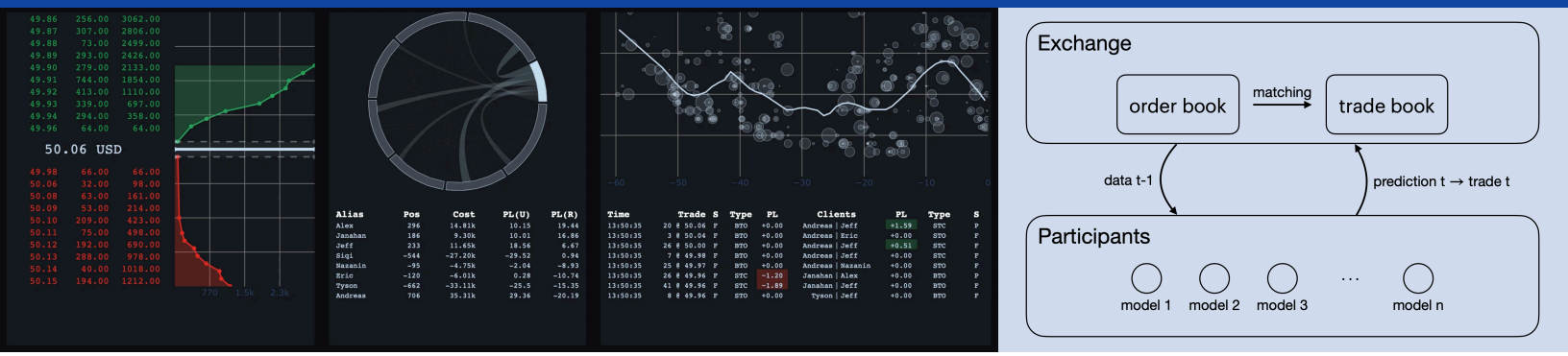


Figure 1: Stock prices in financial markets (**left**) are the result of explicit interactions between competing models (**right**).

Context

Today's equity markets are faster-paced and more complex than ever before. With trading frequencies approaching the nanosecond barrier and financial instruments organized into increasingly elaborate systems of hierarchical layers and nested dependencies, data-driven forecasting and risk control have long been central to quantitative strategies in capital markets. Unlike other machine learning domains, such as computer vision or natural language processing, capital markets are characterized by **competing interactions** between n machine learning models that operate simultaneously on the same data (the "order book"). Each action (an "order placement") alters that shared observable data for everyone, creating a recursive coupling among all models.

Motivation

In financial markets, participants typically interact with an exchange, such as NASDAQ, rather than directly with one another. A simple example is the placement of a buy or sell order for 100 shares of Apple at $125.00. Such orders are often generated by neural networks that process a wide range of features (e.g., historical prices, recent news, economic data) and identify desirable trades based on prior training against a given objective, such as maximizing portfolio returns. The role of the exchange is to match submitted orders against compatible orders from counterparties willing to take the opposite side of the trade, a process known as order book matching. From the perspective of a single participant, this mechanism transforms what are, in effect, interactions with many market participants into interactions with the order book, a single interface representing the collective activity of all counterparties. Well-known challenges in capital markets, including low signal-to-noise ratios and non-stationary dynamics, can be traced back to this superposition of strategies and actions in the order book, making capital markets less interpretable and, ultimately, less predictable. Taking robust actions in such an environment requires novel architectures that reflect and exploit these unique properties.

Project

Instead of modeling future prices as an opaque function of observed features (e.g., based on the information in the order book), this project takes a more holistic and interpretable perspective. Specifically, we view the sequence of stock prices, both past and future, as the **consequence** of n competing machine learning models whose orders are matched in an explicit order book. At training time, we ask what the parameters of these models must be for their orders to produce price matches consistent with actual observed stock prices. The benefits of this approach are threefold: (1) modeling the behaviour of individual participants is easier than modeling their aggregate behaviour in the order book; (2) semantic insight into the strategies of competing models enables the anticipation of future events beyond price dynamics; and (3) leveraging knowledge about other participants' data and architectures enables us to remotely steer the actions of other participants.

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PROJECT MODULE PROPOSAL | AI MASTER

Physical AI: Open-Source Robot Arm with LeRobot

Build robot learning from camera input to a live pick-and-place demo.

Partner: SKZ | Physical AI | Python + PyTorch | Open source

Students build an end-to-end Physical AI demonstrator: an affordable robot arm with camera input, AI-based control, reproducible training data, and two visible demo tasks that make robot learning tangible from week one.

Why this is exciting

- Connect vision, learning, control, and hardware in one visible AI system.
- Move from fixed robot scripts toward learned behavior using LeRobot-style pipelines.
- Create a demo that students, visitors, and company partners can understand immediately.

Core project question

How far can an AI Master project team get with affordable hardware, open-source tools, and a small but well-designed robot-learning dataset?

- Simple task: pick, place, sort, and move objects.
- Stretch task: reproduce small make-and-break structures.

What students will do

- Assemble or adapt the robot arm, camera, and sensing setup.
- Collect demonstrations and visual observations for training.
- Train, evaluate, and document AI control policies.
- Package the result as a reproducible demo and short technical report.

Methods and tools

- Python, PyTorch, OpenCV, Hugging Face ecosystem.
- LeRobot for data, policies, and robot-learning workflow.
- Computer vision for object detection and position estimation.
- Hardware/software integration and reproducible movement pipelines.

Expected outcome

- Working robot-arm demonstrator with one reliable baseline task and one ambitious extension task.
- Documented training and evaluation pipeline that future teams can reproduce and extend.
- Final demo: from camera input to learned action in a real physical setup.

Why this fits the AI Master

- Clear entry point, visible progress, and scalable difficulty from baseline to challenge task.
- Team roles for hardware setup, computer vision, model training, evaluation, and documentation.
- Industry context with SKZ; detailed scope can be tuned during kickoff.

Interested? Contact Prof. Dr. Frank-M. Schleif: frank-michael.schleif@thws.de | personen.fiw.thws.de/schleif/

Sources: Green-INNO project page; SKZ public website; Hugging Face LeRobot documentation

PROJECT MODULE PROPOSAL I AI MASTER

WasteViT: A Foundation Model for Visual Waste Understanding



Give waste streams a visual language for smarter recycling.

Partner: WeSort.AI | Computer Vision | Foundation Models | Real sorting data

Students prototype the first building blocks of a waste-specific vision foundation model: a model that not only labels objects, but starts to understand messy, overlapping, damaged, and unknown waste in real sorting environments.

Why this is exciting

- Waste is visually hard: dirty, deformed, overlapping, partially hidden, and constantly changing.
- The project connects modern AI with circular economy, recycling quality, and industrial deployment.
- WeSort.AI is an award-winning company with installed systems, data, customers, and a working business model.

Core project question

Can a domain-specific vision foundation model learn reusable representations for many waste fractions instead of being narrowly trained for only one material class?

- From single-fraction detection to reusable visual knowledge.
- From "plastic" toward richer semantic explanations.

What students will do

- Explore and curate a representative image corpus from waste-stream data.
- Design a baseline with transfer learning or self-supervised vision models.
- Evaluate generalization across different fractions and visual conditions.
- Document how the model could plug into industrial sorting and analysis pipelines.

Methods and tools

- Vision Transformers, contrastive/self-supervised learning, and transfer learning.
- Python, PyTorch, modern experiment tracking and evaluation.
- Explainability and error analysis for decisions operators can trust.
- Deployment-aware thinking: inference speed and robustness under plant conditions.

Expected outcome

- WasteViT baseline: curated data subset, model, metrics, and visual error analysis.
- Recommendation for the next research sprint: dataset gaps, architecture, evaluation protocol.
- Final demo: what the model sees, what it predicts, and why it matters for recycling.

Why this fits the AI Master

- Strong partner context, real application pressure, and tasks that can scale by team size.
- Team roles for data curation, modeling, evaluation, explainability, and deployment thinking.
- WeSort.AI is a proper company with installed systems and 2024 Deutscher Gruenderpreis recognition.

Interested? Contact Prof. Dr. Frank-M. Schleif: frank-michael.schleif@thws.de | personen.fiw.thws.de/schleif/

Sources: Green-INNO project page; WeSort.AI public website; Deutscher Gruenderpreis 2024 profile

Green-INNO

PROJECT

PERCY

An Agentic Inbox Butler for Microsoft 365

Redirect your mail to an agent that reads, triages and summarises your inbox on a schedule you set — and pings your phone the moment something is genuinely urgent.

Codename after fiction's most dutiful, rule-abiding junior official — the kind who would never, ever let an important memo slip through unnoticed.

Supervisor: Prof. Dr. Ivan P. Yamshchikov (CAIRO — Center for AI and Robotics, FIW) | **Team size:** 3 | **Level:** 2nd-semester MAI | **Language:** English

CONTEXT & MOTIVATION

Academic staff drown in email. The overwhelming majority is low-priority, yet a handful of messages each day are genuinely time-critical — a deadline, a student emergency, a request from leadership. Scanning everything to find the few that matter is a daily tax on attention. THWS runs on **Microsoft 365** (Outlook / Exchange). This project builds an opt-in agentic assistant: the user redirects mail to a dedicated agent mailbox (or grants delegated access), and the agent then (a) produces **regular, configurable summaries** of what arrived and what needs action, and (b) detects **genuinely urgent** messages and pushes a real-time alert to the user's phone via a **Telegram or WhatsApp** bot. The user stays in control of cadence, urgency criteria and channel.

TECHNICAL SCOPE

- **Microsoft 365 integration:** ingest mail via the Microsoft Graph API (OAuth2), or via an Outlook redirect rule into a monitored agent mailbox.
- **Triage & classification:** LLM-based labelling of messages (urgent / action-needed / FYI / noise) under a user-configurable urgency policy.
- **Scheduled summarisation:** digest generation on a cadence the user chooses (e.g. a morning and an evening brief) covering what came in and what needs a reply.
- **Real-time urgent alerts:** push flagged-urgent messages to the user's phone through a Telegram bot (Bot API) or WhatsApp (e.g. via Twilio).
- **User configuration:** a simple settings layer for cadence, urgency rules, alert channel and quiet hours.
- **Data handling:** email is sensitive — the team explicitly addresses where inference runs (cloud vs. self-hosted model) and how content is stored and minimised.

MILESTONES (ALIGNED WITH THE THREE GRADE-CALIBRATION EVENTS)

Poster session	Requirements analysis, Graph/redirect integration design, urgency taxonomy, architecture and chosen stack.
Paper	Methodology of classification + summarisation + alerting; evaluation design and metrics.
Demo	Live agent producing a scheduled digest and delivering a real-time urgent alert to Telegram/ WhatsApp, with user configuration.

DELIVERABLES

A working prototype and source repository; a short technical paper; a poster; a live demo; and an evaluation of urgency detection (precision/recall) and summary quality on an anonymised or synthetic email set.

PREREQUISITES

Python; foundations of ML/NLP; comfort with REST APIs and OAuth2 and with LLM prompting. Familiarity with asynchronous scheduling and bot APIs (Telegram / Twilio) is helpful but can be acquired during the project.

PROJECT

VOGON

The Self-Hosted AI That Fills Out Your Forms So You Never Have To

An on-premise LLM that turns a one-line request into a finished travel request, purchase order and draft email — no cloud, no data leaving the campus.

Codename, with full irony, after the galaxy's most devoted lovers of forms, paperwork and procedure. This one, for once, is on your side.

Supervisor: Prof. Dr. Ivan P. Yamshchikov (CAIRO — Center for AI and Robotics, FIW) | **Team size:** 3 | **Level:** 2nd-semester MAI | **Language:** English

CONTEXT & MOTIVATION

Professors and staff lose a surprising amount of time to repetitive administrative paperwork: travel requests and trip reports (*Dienstreiseantrag*), purchase orders (*Beschaffungsanträge*), and the routine emails that surround them. Each task means re-typing the same facts into rigid forms and templates. A large language model is well suited to this, but university paperwork contains personal and procurement data that should never be sent to an external cloud service. This project asks a focused question: can an **open-weight LLM, hosted entirely on a CAIRO machine**, reliably take a short natural-language request and turn it into a correctly filled form and a ready-to-send draft email — with a human always approving the final result?

REFERENCE CODEBASE & STARTING POINT

The team does not start from a blank page. The German Federal Ministry for Digital Affairs (BMDS) has released **SPARK Workflow** as open source (EUPL 1.2) — a modular, Docker-based suite of AI components for administrative procedures that already implements document information extraction, formal completeness checking, decision/draft generation, and a human-in-the-loop UI with traceable source references. Students study this codebase as a reference architecture and reuse or adapt suitable modules for the university paperwork setting, rather than reinventing them.

Reference codebase (clickable): https://gitlab.opencode.de/bmds/planungs-und-genehmigungsbeschleunigung/spark-workflow/-/blob/main/README.md

TECHNICAL SCOPE

- **Self-hosted inference:** deploy an open-weight LLM on-premise (e.g. via Ollama or vLLM); strictly no external API calls — a GDPR-friendly, data-sovereign setup.
- **Chat interface:** a simple conversational front-end where a user describes a task in plain language ("3-day conference trip to Berlin, 12–14 Oct").
- **Retrieval & structured extraction:** RAG over the relevant THWS forms, templates and policies; map the request to the structured fields of the target form.
- **Form filling & drafting:** produce a filled *Dienstreiseantrag* / purchase-order PDF and a matching draft email; user reviews and approves before anything is sent.
- **Deployment:** containerised (Docker), reproducible, runnable on internal CAIRO hardware.

MILESTONES (ALIGNED WITH THE THREE GRADE-CALIBRATION EVENTS)

Poster session	Problem & requirements analysis, study of SPARK Workflow, chosen architecture and stack, plan.
Paper	Methodology of the extraction + form-filling pipeline; evaluation design (accuracy, time saved, error modes).
Demo	Working prototype filling ≥2 real form types and drafting emails end-to-end, with evaluation results.

DELIVERABLES

A dockerised working prototype and source repository; a short technical paper; a poster; a live demo; and an evaluation on a small set of real (anonymised) forms.

PREREQUISITES

Python; foundations of ML/NLP; comfort with LLM prompting. Familiarity with Docker and basic web front-ends is helpful but can be acquired during the project.

PROJECT

KIWi

KI-Wissensassistenz — AI-Powered Knowledge Assistance for the THWS CampusPortal

A locally hosted, retrieval-augmented LLM that turns complex HISinOne queries into instant, sourced answers — keeping all data on campus.

KIWi stands for KI-Wissensassistenz (AI knowledge assistance). Institutional knowledge shouldn't live only in one expert's head — or leave the building.

Supervisor: Prof. Dr. Ivan P. Yamshchikov (CAIRO — Center for AI and Robotics, FIW) | **Team size:** 3 | **Level:** 2nd-semester MAI | **Language:** English

CONTEXT & MOTIVATION

The THWS CampusPortal (HISinOne) manages central university administration — student registration, module management, finance, and HR. Despite its comprehensive functionality, the system is notoriously complex: configuration options and processes are documented across wikis, training materials, and the tacit knowledge of a handful of experts. Staff lose significant time searching for answers, knowledge walks out when people leave, and external consultancy is expensive. A retrieval-augmented LLM can change this: a system that reads all internal documentation, answers queries instantly with source citations, and never forgets — while keeping every piece of sensitive data within the campus network.

REFERENCE TOOLS & STARTING POINT

The team does not start from a blank page. Mature open-source tooling makes a working prototype achievable within one semester: **Ollama**/**vLLM** for self-hosted LLM inference (Mistral, LLaMA 3, Qwen); **LangChain** or **LlamaIndex** for the RAG pipeline (chunking, retrieval, prompting); **ChromaDB** or **pgvector** as the vector store. Students select and integrate appropriate components rather than reinventing them. Domain knowledge, documentation, and user requirements are provided by the **THWS HISinOne team**, who serve as the operational partner throughout.

TECHNICAL SCOPE

- **Local LLM deployment:** self-hosted open-weight model (Ollama/vLLM); no external API calls — GDPR-compliant and data-sovereign.
- **Knowledge ingestion:** automated pipeline to load and index HISinOne wikis, training materials and process documentation into a vector store.
- **RAG pipeline:** semantic retrieval over the internal knowledge base combined with LLM generation; responses include source references.
- **Chat interface:** conversational front-end where a user types a plain-language query and receives a cited, ready-to-act-on answer.
- **Quality assurance & deployment:** evaluation of accuracy, relevance and hallucination rate; human-in-the-loop validation; Dockerised and reproducible.

MILESTONES (ALIGNED WITH THE THREE GRADE-CALIBRATION EVENTS)

Poster session	Requirements analysis with the HISinOne team; knowledge audit; chosen stack and architecture; pipeline plan.
Paper	RAG pipeline design and knowledge base construction; evaluation framework (accuracy, relevance, hallucination rate).
Demo	Working prototype handling real (anonymised) CampusPortal queries end-to-end, with evaluation results.

DELIVERABLES

A Dockerised prototype and source repository; a short technical paper; a poster; a live demo; and an evaluation on a representative set of real (anonymised) queries.

PREREQUISITES

Python; NLP and LLM foundations; interest in RAG systems and knowledge management; comfort with LLM inference frameworks (Ollama, vLLM) or vector stores is a strong plus. This project requires close collaboration with the THWS HISinOne team — the ability to translate operational requirements into technical decisions is essential. Docker is helpful but can be learned on the job.

Technical University of Applied Sciences Würzburg-Schweinfurt
Faculty of Computer Science and Business Information Systems
Study Program: Master Artificial Intelligence (MAI) · Winter Semester 2026/2027

Supervisors: Prof. Dr. Frank Deinzer, M.Sc. Max Werner
Contact: frank.deinzer@thws.de

Project Module Topic:
Data-Driven Pedestrian Dead Reckoning and Activity Recognition

Pedestrian Dead Reckoning (PDR) is a method to track the relative motion of a person based on sensor data, e.g. accelerometer or gyroscope data [1]. PDR typically involves step detection and estimation of step length and heading. Based on the pedestrian's previous state, consisting of location and heading, this information can be used to estimate a new state.

Idea

The goal of the proposed project is to replace an existing rule-based PDR model by a data-driven solution. The approach is set to be integrated in a smartphone-based indoor positioning system that helps pedestrians to find their location and navigate inside buildings [2]. It is supposed to build upon an existing Master's thesis is solely concerned with step detection [3]. In this project, however, a model should be developed that is trained end-to-end on full transitions between locations. An interesting addition to the PDR model would be activity recognition, which could provide useful information for the positioning system regarding how it should react to changes.

Main Tasks

- Create a dataset of walking people carrying smartphones that record sensor data. Accurate ground truth positions may be provided by an external ultra-wideband positioning system. All hardware will be provided by us.
- Train a PDR model based on the dataset. Based on a sequence of smartphone sensor data, it should estimate the position and heading change of the pedestrian. You can feel free to use advanced machine learning or deep learning approaches.
- Train an activity recognition that also classifies whether the pedestrian is currently standing, walking, running or doing something else.
- Write a research paper where you describe your dataset and model. A scientific evaluation could contain a comparison to other state-of-the-art approaches [4], [5].

References

[1] S. Beauregard and H. Haas, "Pedestrian dead reckoning: A basis for personal positioning," *WPNC'06*, 2006.
[2] T. Fetzer, F. Ebner, M. Bullmann, F. Deinzer, and M. Grzegorzek, "Smartphone-based indoor localization within a 13th century historic building," *Sensors*, vol. 18, no. 12, 2018.
[3] M. Z. S. Sarwar, "Step detection," Master's thesis, THWS, 2023.
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Probabilistic Deep Learning Group PDLG

NeuroKineX

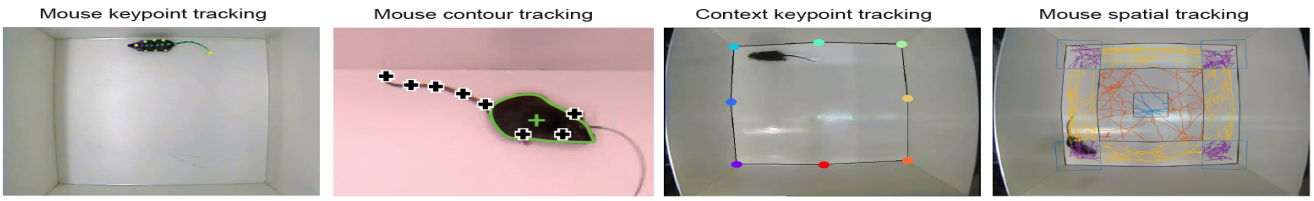
Neurological Behavioural Kinematic Extrapolation via Generative Modelling

IN COLLABORATION WITH

Universitätsklinikum Würzburg – Neurologische Klinik

Clinical partner and provider of the behavioural dataset.

Uniklinikum Würzburg



Pose-tracking data: each mouse recording yields keypoints, contour and spatial trajectory — the raw material for NeuroKineX.

Why it matters

Parkinson's disease, dystonia and spastic paraplegia progressively rob people of ordinary movement — tremor, spasms and rigidity erode independence and quality of life, often for decades without a cure. Studying these disorders in detail requires observing and manipulating movement under controlled conditions impossible in human patients. Mouse models fill this gap: the same conditions can be induced and perturbed (e.g. via deep brain stimulation or drug treatment), and the animal's movement — how it walks, grooms and rears — is the measurable readout.

The project

Can a generative model trained on diverse movement profiles extrapolate to disease–intervention combinations it has never seen?

You will work with a dataset provided by the Neurologische Klinik at UKW containing pose-tracking time series (skeletal keypoints) of freely moving mice, each labelled with disease model, intervention, and behavioural state. Your goal is to build a conditional generative model — diffusion-based or via flow matching — that generates realistic movement sequences for **label combinations absent from training**: for instance, the effect of deep brain stimulation on a dystonic mouse when no such recording exists. We suggest exploring inference-time steering strategies that allow compositional extrapolation without retraining for every new condition. You will evaluate generated sequences against held-out real recordings using quantitative metrics and behavioural classifiers. The expected deliverable is a documented, reproducible codebase.

Why pick this project

This project is an opportunity to work at the intersection of state-of-the-art deep generative modelling and real clinical research — contributing to how neurological movement disorders are studied and ultimately treated. You will work with real data provided by the Neurologische Klinik at UKW, on an open scientific question that genuinely matters to the clinicians we collaborate with. The *PDLG group* works on exactly the methods this project builds on: diffusion models, flow matching, and conditional generation, and can provide expert guidance. For a strong team, a peer-reviewed publication is a highly desirable outcome.

Methods & starting points

- **Generative backbone:** Continuous-time modeling via Flow Matching (Lipman et al., ICLR 2023).
- **Classifier-free guidance:** Standard conditional generation baseline (Ho & Salimans, 2022).
- **Inference-time steering:** Reward-guided sampling for diffusion models (Singhal et al., 2025).
- **Behavioral context:** Trajectory segmentation from keypoint data (Weinreb et al., Nature Methods 2024).

YOU ARE A GOOD FIT IF...

You are curious and inventive — generative modelling here means building and training architectures, not prompting pretrained models. You work at the implementation level: reading papers and turning methods into running code. You are solid in Python and PyTorch, fluent with standard code development practices (version control, containerisation, reproducible workflows), and mathematically mature — diffusion models and flow matching are not light on mathematics (probability theory, stochastic processes, continuous time dynamics). Co-attending the parallel *Deep Unsupervised Learning* course is strongly recommended!

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Probabilistic Deep Learning Group PDLG

FusionAD

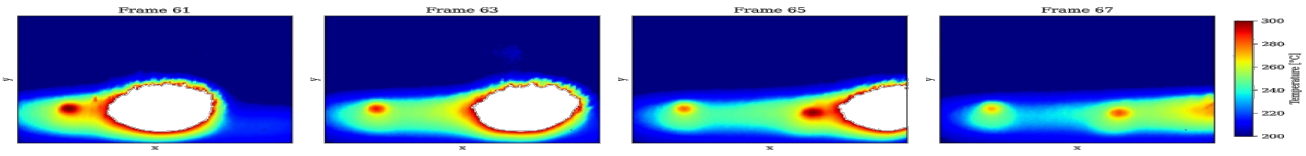
Defect Detection in Laser Powder Bed Fusion via Thermal Sequence Modelling

IN COLLABORATION WITH

Technologietransferzentrum Main-Spessart (TTZ-M) – THWS

Research partner, provider of active thermography measurement data and domain expertise.

TTZ-M
ttz-m.thws.de



Thermal sequence frames: the laser sweeps across the surface and the cooling response is recorded — subsurface defects disturb local heat dissipation and appear as anomalous patterns.

Why it matters

Metal parts produced by laser powder bed fusion (a metal 3D printing process) are used in aerospace, medical implants and automotive — industries where a hidden subsurface defect can cause structural failure. Active thermography offers a non-invasive window into the material, but the thermal response is a complex spatiotemporal signal: distinguishing real defect signatures from laser artefacts and measurement noise automatically and reliably is an unsolved challenge.

The project

Can a model learn to distinguish normal from anomalous thermal responses to reliably localise subsurface defects?

During production, a laser sweeps across the component surface while a thermal video is recorded. Because the exact laser trajectory is known, the data can be transformed into a laser-centered coordinate system that stabilizes the thermal cooling dynamics and reduces geometric variability. The dataset, provided by TTZ-M, includes thermal sequences across multiple components along with ground-truth defect labels. Your task is to design a framework that maps these spatiotemporal inputs to a spatial defect map.

From this starting point, you can explore supervised spatiotemporal localisation with a U-Net/Transformer hybrid, or unsupervised anomaly detection based on reconstruction or prediction errors — comparing the model's expected cooling behaviour against what was actually observed. The key challenge throughout is to robustly distinguish true subsurface defects from laser artefacts and measurement noise.

Why pick this project

Advanced machine learning and engineering are converging fast, but researchers who operate fluently at this boundary — understanding both deep learning methodology and physical measurement processes — are highly sought after. This project gives you exactly that experience: working with real data on a problem that directly affects manufacturing quality and safety. Guided by the *PDLG group* in collaboration with the partner research institute, you will learn how to combine physics-aware preprocessing, modern sequence models, and robust anomaly detection on a challenging industrial dataset. A paper would be a highly desirable outcome, and the project may also serve as a basis for longer-term research collaboration that could be continued within a master's thesis.

Methods & starting points

- **Sequence backbone:** Factorised spatiotemporal attention for thermal video modelling (Bertasius et al., ICML 2021).
- **Supervised localisation:** U-Net/Transformer hybrid for spatiotemporal segmentation (Chen et al., arXiv 2021).
- **Sequence-level anomaly detection:** Out-Of-Distribution via likelihood scoring (Shchur et al., NeurIPS 2021).
- **Active thermography defect detection:** Spatiotemporal DL, lock-in thermography (Schmid et al., NDT&E Int. 2024).

YOU ARE A GOOD FIT IF...

You are curious about how physical measurement processes work and excited to apply machine learning to structured, physics-governed data — you approach experimentation systematically and are not deterred by messy real-world data. You work at the implementation level: reading papers and turning methods into running code, building and training models rather than prompting or orchestrating pretrained LLMs. Some background in signal processing or time series analysis is helpful, as the project involves non-trivial preprocessing before any modelling begins. You are solid in Python and PyTorch, comfortable with probability theory, and fluent with standard code development practices (version control, containerisation, reproducible workflows). Co-attending the parallel *Deep Unsupervised Learning* course is recommended — VAEs and related generative models are directly applicable here.

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Center for Artificial Intelligence

Prof. Dr. Magda Gregorová
Probabilistic Deep Learning Group PDLG

CellFoundry

Foundation Models for Cell Instance Segmentation

IN COLLABORATION WITH

Institut für Medizintechnik Schweinfurt (IMES) – THWS

Research partner, provider of microscopy imaging data and biomedical domain expertise.

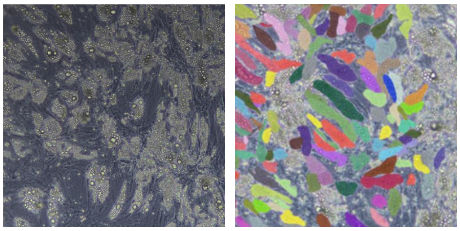
IMES
thws.de/imes

Why it matters

Adipose tissue biology is central to understanding obesity, type 2 diabetes and metabolic syndrome. Adipocyte size and count are the primary morphological readouts, but manual annotation of microscopy images is not scalable to the volumes required for research. Human brightfield images present the hardest case: cell boundaries are diffuse and frequently overlapping, contrast is low, and cell morphology varies substantially with differentiation state — making automated instance segmentation an unsolved practical challenge.

The project

Does the text-prompt capability of SAM3-era foundation models improve adipocyte instance segmentation in human brightfield microscopy compared to SAM2-era microscopy-specific models?



Left: Human tissue brightfield. *Right:* Partial annotation.

Two generations offer complementary strengths: **SAM2-era models** (Micro-SAM, CellSAM) are fine-tuned for microscopy using point or box prompts; **SAM3-era models** (Medical SAM3, SAM3-Adapter) accept concept prompts — short noun phrases like *"round fat cell"* — where visual cues alone are ambiguous.

Your task is to benchmark these model families systematically. Starting from zero-shot evaluation, you will fine-tune each candidate on annotated IMES data — using Micro-SAM/torch-em for SAM2-era models and SAM3-Adapter for SAM3-era models — and compare against the baseline performance of the existing IMES segmentation tool (to be provided by IMES at the outset of the project). A non-trivial part of the work lies before any model is run: raw brightfield images require illumination correction, contrast enhancement (e.g. CLAHE) and intensity normalisation, and careful data augmentation — geometric transformations, intensity jitter, elastic deformations — is critical at the fine-tuning stage. Iterative pseudo-labelling on unannotated IMES images is a secondary direction for a strong team. The expected deliverable is a documented, reproducible evaluation pipeline and a recommended fine-tuned model.

Why pick this project

Foundation models are transforming biomedical image analysis, and text-promptable segmentation is one of the most active frontiers. Whether SAM3-era concept prompts genuinely help for ambiguous microscopy data is an open question — your results will be among the first to answer it. You will work at the intersection of state-of-the-art computer vision and a real clinical-research problem, on data that matters to the researchers at IMES. The experience you gain — benchmarking foundation models rigorously, adapting research codebases, understanding when preprocessing beats model choice — is directly valuable in medical imaging industry and research alike.

Methods & starting points

- **Micro-SAM:** SAM fine-tuned for microscopy, with training infrastructure (Archit et al., Nature Methods 2024).
- **CellSAM:** Foundation model for cell segmentation across imaging modalities (Israel et al., Nature Methods 2025).
- **Medical SAM3:** SAM3 fully fine-tuned on medical imaging data with text prompts (Tian et al., arXiv 2026).
- **SAM3-Adapter:** Lightweight adapter fine-tuning of SAM3 (Chen et al., arXiv 2025).

YOU ARE A GOOD FIT IF...

You are curious about biomedical imaging and excited to work with state-of-the-art foundation models — not as black-box APIs, but at the fine-tuning and evaluation level. You pick up unfamiliar codebases quickly, reading documentation and source code to understand how a model works and what needs to change for your data. You design experiments systematically, vary one thing at a time, and document your findings carefully. You are solid in Python and PyTorch and fluent with standard code development practices (version control, containerisation, reproducible workflows). Some familiarity with image processing is helpful. You communicate clearly: this project requires translating domain requirements from biomedical researchers at IMES into concrete data analysis tasks, and reporting results back in terms that are meaningful to non-ML experts.

Knowledge Distillation for Vision Transformers

Motivation

Vision Transformers achieve excellent accuracy but are often too computationally expensive for edge devices. Knowledge distillation can transfer knowledge from large models to compact ones, enabling efficient deployment while maintaining high performance

Expected Work

- Study Vision Transformers and knowledge distillation techniques
- Implement and evaluate a teacher–student architecture
- Train a compact Vision Transformer using distillation
- Compare accuracy, model size, and inference speed
- Deploy and evaluate the model on a target edge device
- Analyze the trade-off between efficiency and performance

Reference: https://openaccess.thecvf.com/content/CVPR2024/papers/Wang_KD-DETR_Knowledge_Distillation_for_Detection_Transformer_with_Consistent_Distillation_Points_CVPR_2024_paper.pdf

Crowdsourced GNSS-Interference Emitter Localization using Graph Neural Networks with Scalable Receiver Networks

Motivation

Interference in wireless positioning systems such as Global Navigation Satellite Systems (GNSS) can significantly reduce reliability in critical applications. Traditional localization approaches often depend on centralized infrastructure or ideal signal assumptions, limiting their effectiveness in realistic, noisy, and partially observed environments.

Expected Work

Develop and evaluate a crowdsourced jammer localization framework that:

- uses graph-based learning for spatial inference that scales with the number of receivers
- remains robust under varying signal conditions and noise levels
- demonstrates that spatial relationships in distributed measurements are sufficient for accurate localization even under changing signal power conditions

Reference: D. Herzalla, W. T. Lunardi, and M. Andreoni. “Graph Neural Networks for Jamming Source Localization”, In ECML PKDD, June 2025.

Artificial intelligence-enabled robots for 3D-scanning of teeth and gums

Supervisor: Pascal Meissner (in collaboration with University Hospital Wuerzburg)
Team Size: 3

Technologies involved: vision, motion planning, 3D-reconstruction, arm-type robots

Background



Robotics has the potential to play an important part in the solution of major concerns which affect us all, such as aging populations & healthcare, labor shortages & manufacturing, sustainable agriculture & food security, and education & social inclusion. We at THWS investigate how artificial intelligence in combination with intra-oral scanning devices can leverage the use of robots to ensure fast, safe, and gentle diagnostics for dentist patients whenever medical staff is not available.

Objectives

The overarching aim of this project is to demonstrate the potential of artificial intelligence methods in enabling robots, rather than humans, to capture complete 3D-scans of patients' mouths. The aim is to combine state-of-the-art motion planning for robot arms with the 3D-reconstruction technology of a commercial intra-oral scanner in order to create a first prototype where a robot maneuvers a scanner autonomously.



- Conducting a *literature survey*, requiring *familiarisation* with the prominent algorithms and systems for next-best-view estimation, stereo vision, and point cloud registration. This stage will include setting up a development environment and studying related tutorials.
- *Implementing* a select number of the above algorithms, *integrating* them for a robot arm to scan the mouth of a phantom head. This stage will include testing in simulation or reality.
- *Experimentally* quantifying and comparing the performance and limitations of the implemented algorithms and robot/scanner setup.

Requirements

Basic knowledge of programming (C/C++, Python) and Linux systems is required for this project. No knowledge of hardware or control required!

Interested? Contact: pascal.meissner@thws.de

Simulation-Driven Deformable Object Manipulation

Supervisors: Prof. Dr. Pascal Meissner, Tamara Bergerhoff
Team Size: 3

Technologies involved: vision, robot manipulation, physics simulation, machine learning

Background

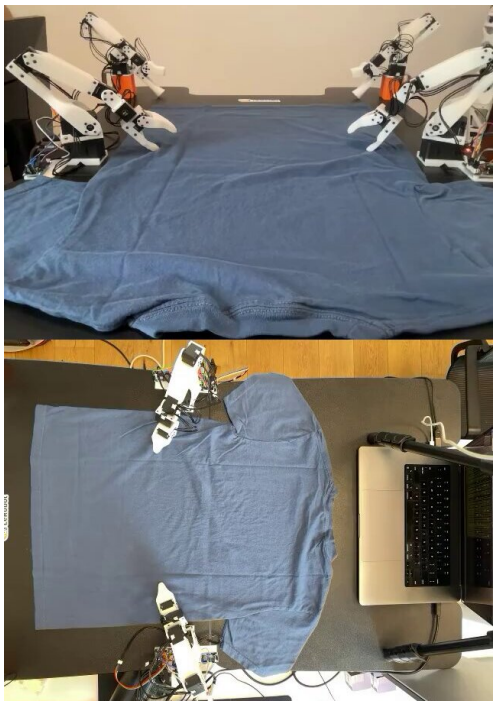


Manipulating deformable objects such as towels, clothes, blankets, or bags remains one of the most challenging problems in robotics due to their complex and unpredictable behaviour. To address these challenges, THWS is establishing a student robotics team to participate in the LeHome Challenge. The project offers the opportunity to contribute to the development of autonomous manipulation algorithms, evaluate them in simulation, and prepare them for deployment on a real robot platform.

Objectives

The aim of this project is to develop and evaluate methods for autonomous manipulation of deformable objects for participation in the LeHome Challenge. The goal is to build a competitive system that can be evaluated both in simulation and on the real robot platform. Deformable object manipulation requires robust perception, grasp planning, and manipulation strategies due to the highly variable shape and physical behaviour of these objects.

- Conducting a *literature survey*, requiring *familiarisation* with deformable object manipulation, robot perception, and physics simulation. This stage will include setting up the development environment, studying relevant software frameworks, and analysing state-of-the-art approaches.
- *Developing* algorithms for deformable objects in a simulation and integrating them into the robot software stack. Depending on the project focus, this may include object detection, grasp generation, motion planning, or learning-based manipulation strategies.
- *Experimentally* evaluating the developed methods and investigating their transfer to the real robot platform for participation in the LeHome Challenge.



Requirements

Basic knowledge of programming (C/C++, Python) and Linux systems is required for this project. An interest in robotics hardware and hands-on experimentation is highly recommended.